

Mechanical Engineering

Course Notes – Topic-wise Overview

Page 1: Engineering Fundamentals

Topics covered:

- Engineering mechanics
- Forces, moments and equilibrium
- Stress and strain
- Strength of materials

Learning Focus

This section introduces the important concepts of engineering fundamentals. Students can use these topics as a starting point for lectures, assignments, revision and further study. Practical examples and laboratory activities can be added according to the college syllabus.

Page 2: Thermal & Fluid Systems

Topics covered:

- Thermodynamics
- Heat, work and energy
- Fluid mechanics
- Heat transfer

Learning Focus

This section introduces the important concepts of thermal & fluid systems. Students can use these topics as a starting point for lectures, assignments, revision and further study. Practical examples and laboratory activities can be added according to the college syllabus.

Page 3: Manufacturing & Design

Topics covered:

- Casting, machining and welding
- Forming and additive manufacturing
- Machine design
- Engineering drawing and CAD

Learning Focus

This section introduces the important concepts of manufacturing & design. Students can use these topics as a starting point for lectures, assignments, revision and further study. Practical examples and laboratory activities can be added according to the college syllabus.

Page 4: Production & Applications

Topics covered:

- Production planning
- Quality control
- Maintenance engineering
- Automobile and mechanical systems

Learning Focus

This section introduces the important concepts of production & applications. Students can use these topics as a starting point for lectures, assignments, revision and further study. Practical examples and laboratory activities can be added according to the college syllabus.